SAY WATT ROBOTICS · FIELD LOG

Week 2.5 — Cardboard, Then Plastic, Then Brass — The Buoyancy Engine Takes Shape

Sunday, October 18, 2026 · 3:00 – 5:00 PM · 8 students

Eight of us at 2.5 — Trenton, Liam, Luke, Adrian, Tyler, Rebecca, Rhylan, and Sam — from 3:00 to 5:00, and the buoyancy engine — the part of the float worth 20 of its 70 points — started to look like a machine. Electrical finally soldered the 12-volt to 5-volt buck converter, closing the joint that the 18-gauge wire wouldn't fit back at 2.3. Then they reconnected the ESP32 to a TMC2209, a quiet stepper motor driver that turns pulses from the microcontroller into precise steps, reloaded the software, and tested the new stepper motor on a 240-millimeter lead screw. A stepper suits this job because steps are countable: count them and you know exactly how far the plungers moved and how much water the float has taken on. One detail from the notes is worth underlining — last season's electrical schematic is what made rebuilding the controller quick. Next to the bench sat the float's wiring diagram, drawn by hand: a 12-volt battery through a single fuse, the buck converter for the ESP32, the motor driver and stepper on the 12-volt side with a 470-microfarad capacitor across the supply, and a micro SD card, a pressure sensor, and a light sensor hanging off the ESP32. Mechanical finalized the end cap design and did the math for its seal: they calculated the depth of the groove the O-ring sits in and settled on a 25 percent squeeze. They also built a working CAD model of the plunger system, and the syringe plate showed its whole iteration history on the table in three pieces — a cardboard cutout, a first 3D print, and a second print with a brass lead-screw nut pressed into the center. The search for a new marketing lead continued.

8	25%	240 mm	3
STUDENTS AT THE SESSION	O-RING SQUEEZE THE END CAP SEAL IS DESIGNED FOR	LEAD SCREW ON THE NEW STEPPER MOTOR	GENERATIONS OF THE SYRINGE PLATE — CARDBOARD, PRINT, PRINT WITH BRASS NUT

WHAT WE LEARNED

- A stepper motor moves in exact steps. Count them and you know how far the plungers traveled, which is how the float knows how much water it is holding without a separate sensor.
- An O-ring seals by being squeezed. The groove it sits in has to be cut to a depth that compresses the ring by a set fraction of its thickness — too little and it leaks, too much and it can't seat. The team designed theirs for 25 percent.
- Last season's paperwork paid off. The old electrical schematic is what made rebuilding the buoyancy controller fast — which is the whole reason you draw schematics in the first place.
- A stepper driver wants a big capacitor across its motor supply. Motors throw voltage spikes back into the supply when they start and stop, and the capacitor soaks them up before they reach the driver.
- A lead screw multiplies force. A small motor turning a screw can push and pull syringe plungers against the water pressure at four meters, and a fine-pitch screw holds its position when the motor stops.
- Iterate in the cheapest material first. Cardboard tests the layout for free, the first print tests the shape, and the second print adds the hardware.

DECISIONS MADE

- The buoyancy engine runs on a stepper motor through a TMC2209 driver, driving a 240 mm lead screw. Steps are countable, so the plunger position — and the water the float is carrying — is known, not guessed.
- The end cap seals on an O-ring, with the groove depth calculated for 25 percent squeeze.
- The 12 V to 5 V buck converter is soldered, closing an item that had been open since 2.3.
- The float's power system is on paper: a 12 V battery through a single fuse, a 5 V buck converter for the ESP32, and the motor driver on the 12 V side with a 470 µF capacitor across it.
- The controller gets its own circuit board. The next electrical step is a PCB, starting with the ESP32 drawn properly in KiCad.

ACTION ITEMS

Create the profiler's circuit board (PCB)
Electrical group · by 2.6

Find the ESP32 KiCad file, or redraw the ESP32 in KiCad
Electrical group · by 2.6

STUDENT LEARNING OUTCOMES

- Soldered the 12 V to 5 V buck converter.
- Reconnected the ESP32 and TMC2209 stepper driver, reloaded the controller software, and tested a new stepper motor on a 240 mm lead screw.
- Calculated the O-ring groove depth for the end cap, designing for 25 percent squeeze.
- Finalized the end cap design and built a working CAD model of the plunger system.
- Drew the profiler's power and signal diagram: battery, fuse, buck converter, ESP32, motor driver, stepper motor, SD card, and pressure and light sensors.
- Iterated the syringe plate from cardboard to two 3D-printed versions, the second carrying a brass lead-screw nut.

LOOKING AHEAD

Start the profiler's circuit board in KiCad, beginning with the ESP32. The hand-drawn wiring diagram is the starting point for the float's System Integration Diagram, and the data packets have to carry depth, water temperature, and light with a timestamp — the format MATE scores. The full 2027 competition manual is expected in November; when it lands, every number from the preview mission gets checked against the final rules.

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